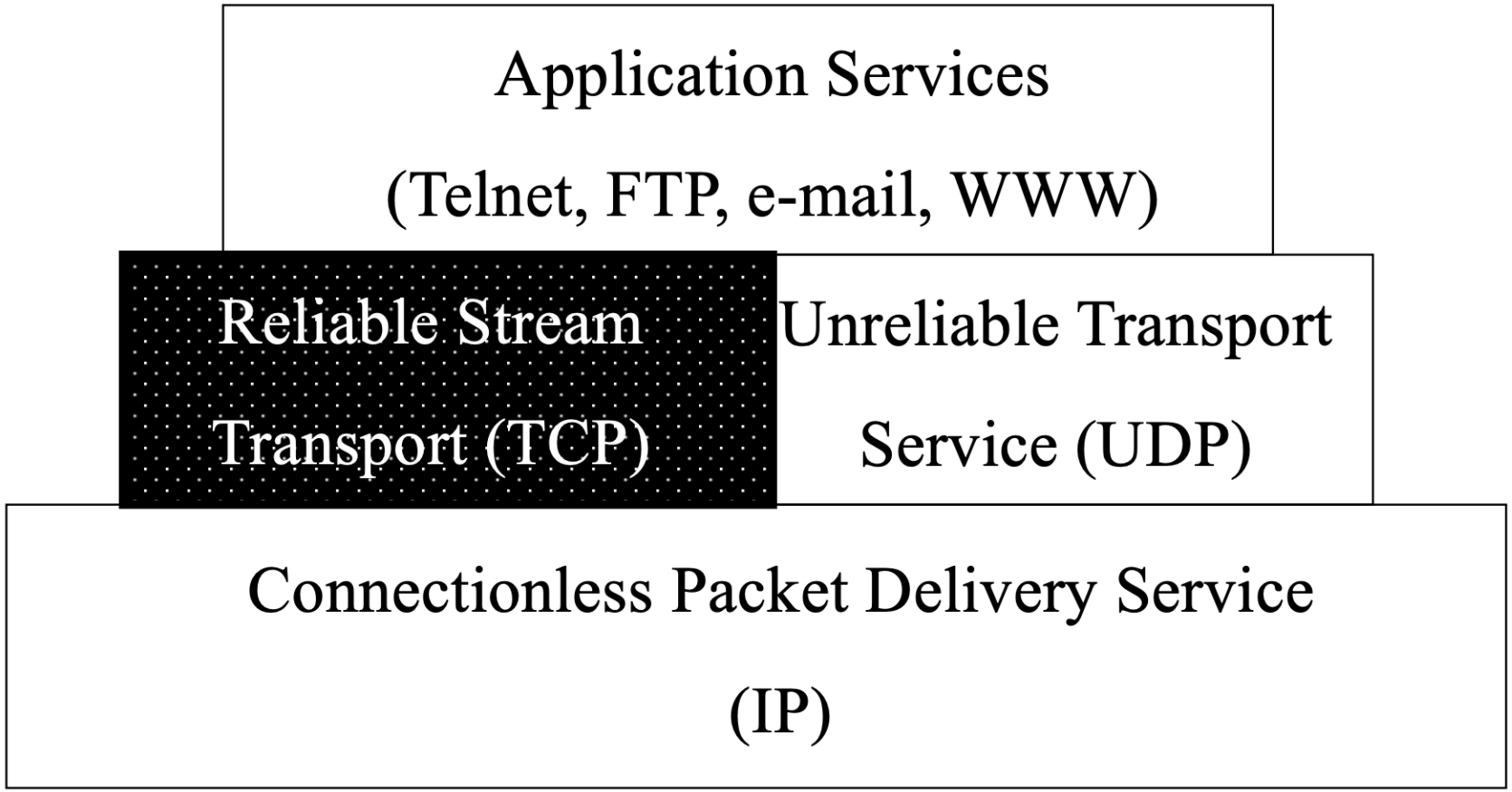
TCP

DaeHun Nyang@Ewha WU

The Transmission Control Protocol (TCP)



The Transmission Control Protocol (TCP)

- TCP is a protocol that specifies:
  - How to distinguish among multiple destinations on a given machine
  - How to initiate and terminate a stream transfer
  - Format of the data and acknowledgments that two computers exchange to achieve a reliable transfer
  - Procedures the computers use to ensure that the data arrives correctly

TCP Acknowledgment (ACK)

- TCP uses ACKs to confirm successful data reception.
- Each byte in TCP has a sequence number.
- The receiver sends back an ACK indicating the next expected byte.
- Example:
  - If sender transmits bytes 100–199, receiver replies with:
  - → ACK = 200 (i.e., "I received up to 199, send from 200.")

TCP Retransmission Basics

- TCP ensures reliability by retransmitting lost packets.
- Retransmission is triggered when:
  1. ACK is not received within timeout → Timeout retransmission
  2. Duplicate ACKs are received → Fast retransmission

Duplicate ACK and Fast Retransmit

- If a packet is lost but later ones arrive:
  - Receiver keeps sending duplicate ACKs.
  - After 3 duplicate ACKs, sender does a Fast Retransmit.
- Benefit: Faster recovery than waiting for timeout.

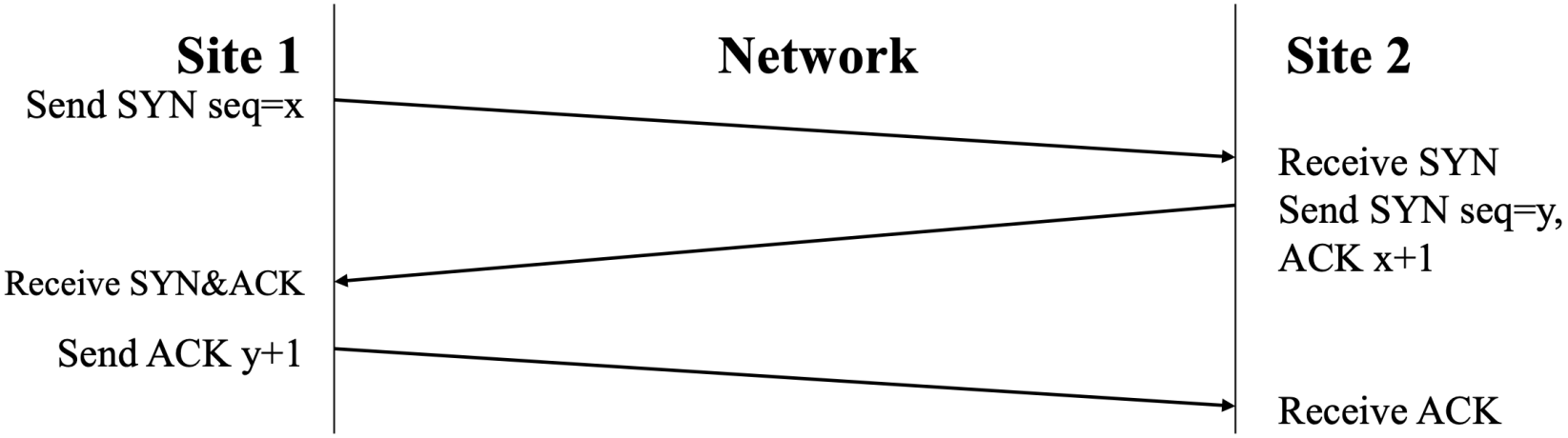
Normal TCP Flow Example

1. SYN Seq=0
2. SYN-ACK Seq=0 Ack=1
3. ACK Seq=1 Ack=1
4. Data Seq=1 Len=1000
5. ACK Ack=1001

- Receiver confirms receipt by sending ACK with next expected sequence number.

Establishing a TCP Connection

- The 3-way handshake
  - Guarantee that both sides are ready for connection
  - Allows both sides to agree on initial sequence numbers



Packet Loss and Fast Retransmit Example

- 10 Data (Seq=1001) ← lost
- 11 Data (Seq=2001)

- 12 ACK=1001 ← Duplicate #1
- 13 ACK=1001 ← Duplicate #2
- 14 ACK=1001 ← Duplicate #3 → Fast retransmit

- 15 Retransmit Seq=1001
- 16 ACK=3001

- Sender detects loss via duplicate ACKs and retransmits immediately.

TCP Window

- The TCP window is a key element of flow control in TCP.
- It helps control the flow of data between sender and receiver.
- Ensures that the sender does not overwhelm the receiver's buffer capacity.

Role of TCP Window

1. Flow Control: Limits the amount of data the sender can transmit based on the receiver's buffer.
2. Window Size: Represents the amount of data the receiver can handle.
3. Dynamic Adjustment: The window size adjusts dynamically based on the receiver's available space.

Sliding Window Mechanism

1. Sender transmits data up to the window size.
2. Receiver processes data and sends an ACK.
3. Sender slides the window based on the ACK and continues sending data.

- This allows continuous and efficient data transmission.

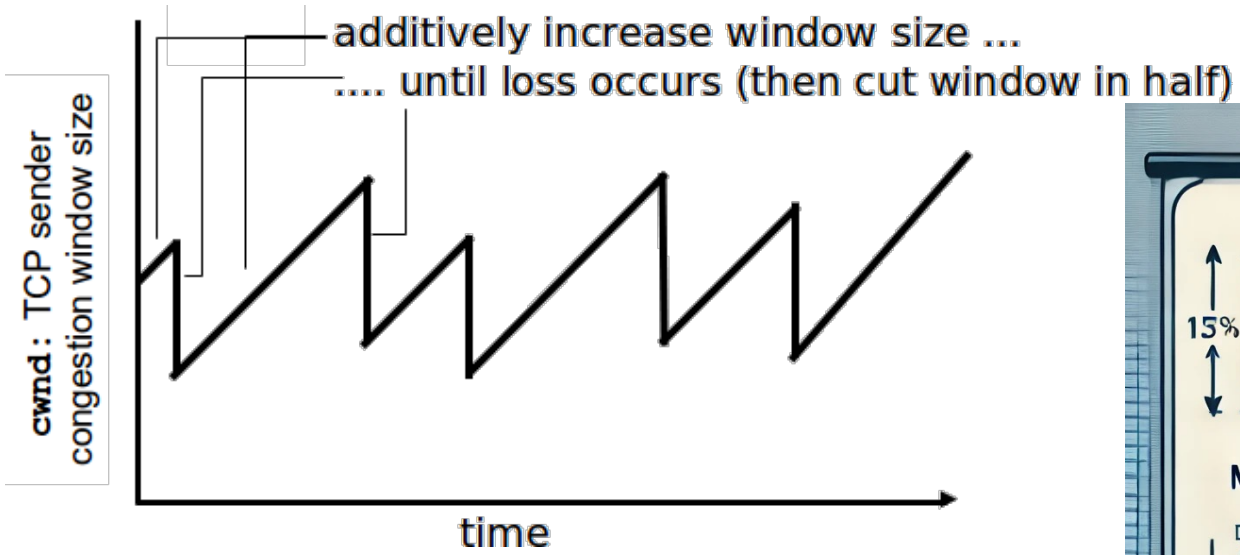
Window Overflow

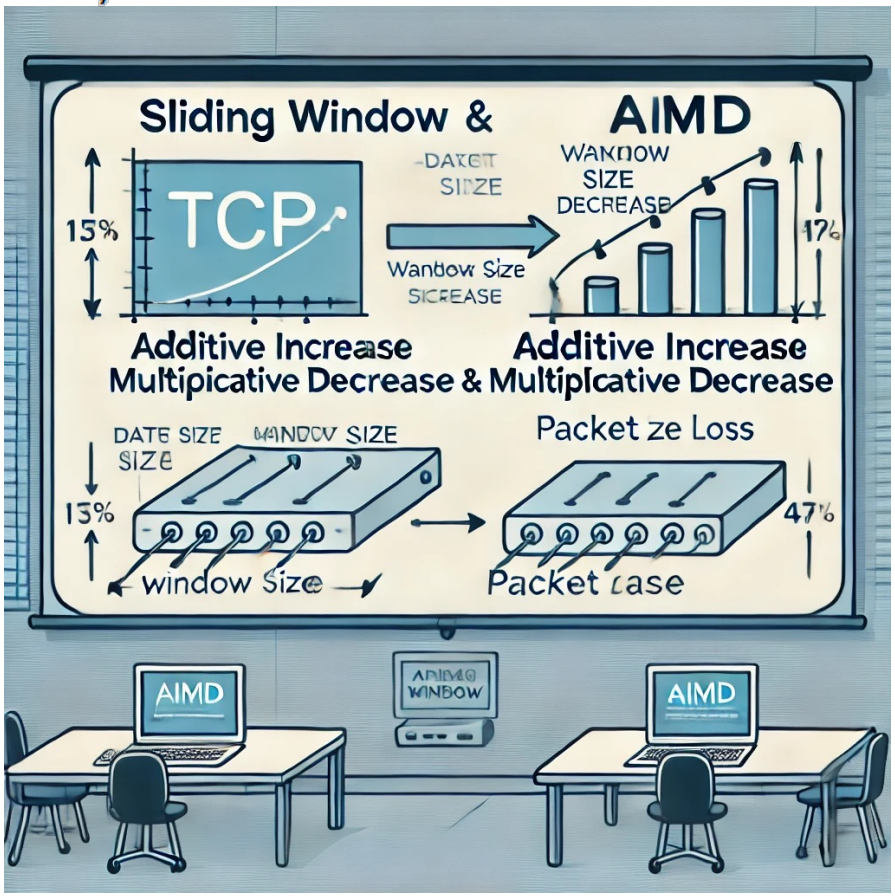
1. Occurs when the receiver's buffer is full.
2. The receiver sends a zero window size to the sender.
3. This prevents the sender from sending more data until space is available.

- Helps prevent network congestion and buffer overflow.

TCP Sliding window for flow/congestion control

- Additive Increase Multiplicative Decrease

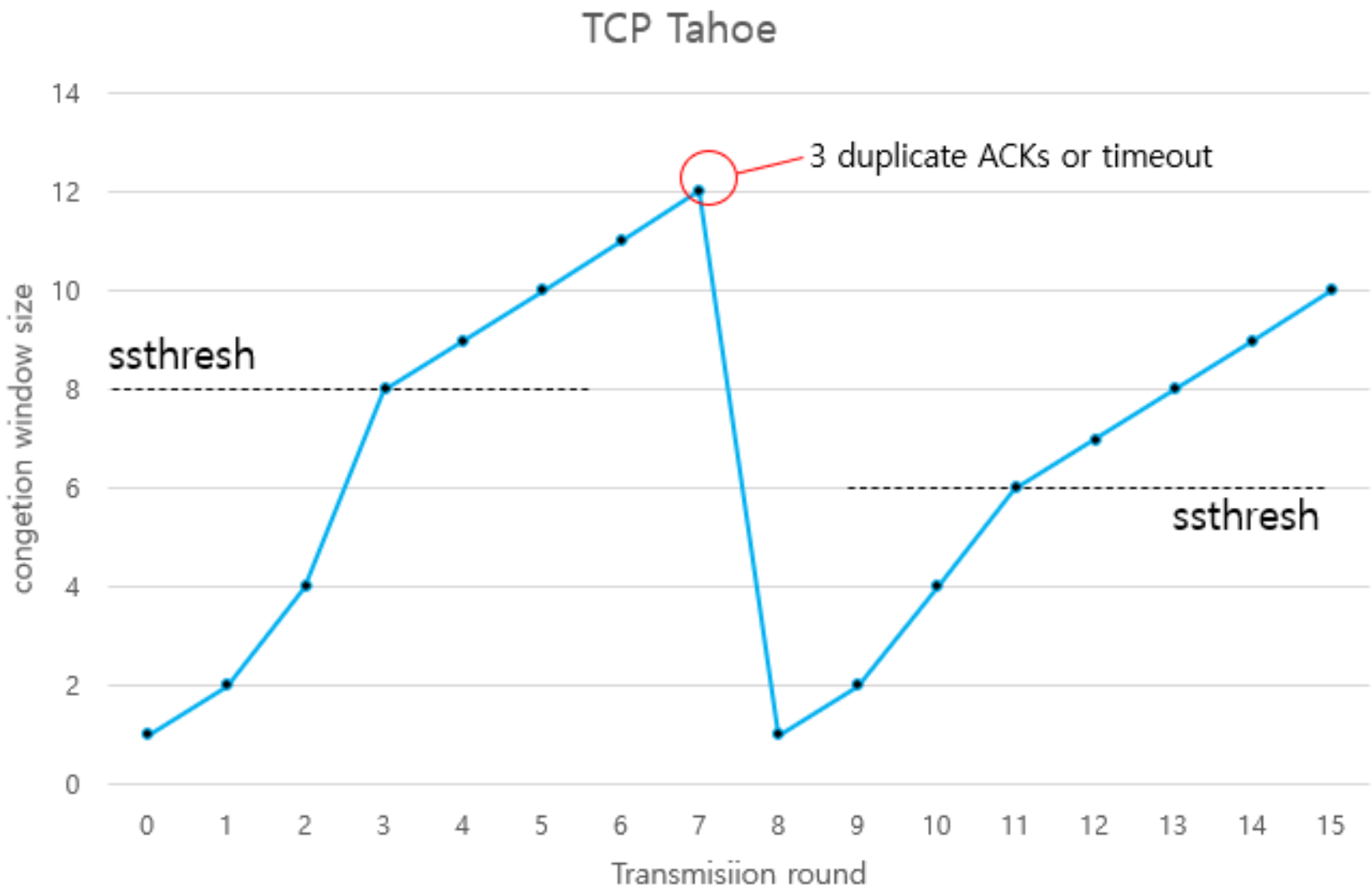




TCP Tahoe

TCP Tahoe initially uses Slow Start, and then switches to the AIMD (Additive Increase Multiplicative Decrease) approach upon reaching a threshold.

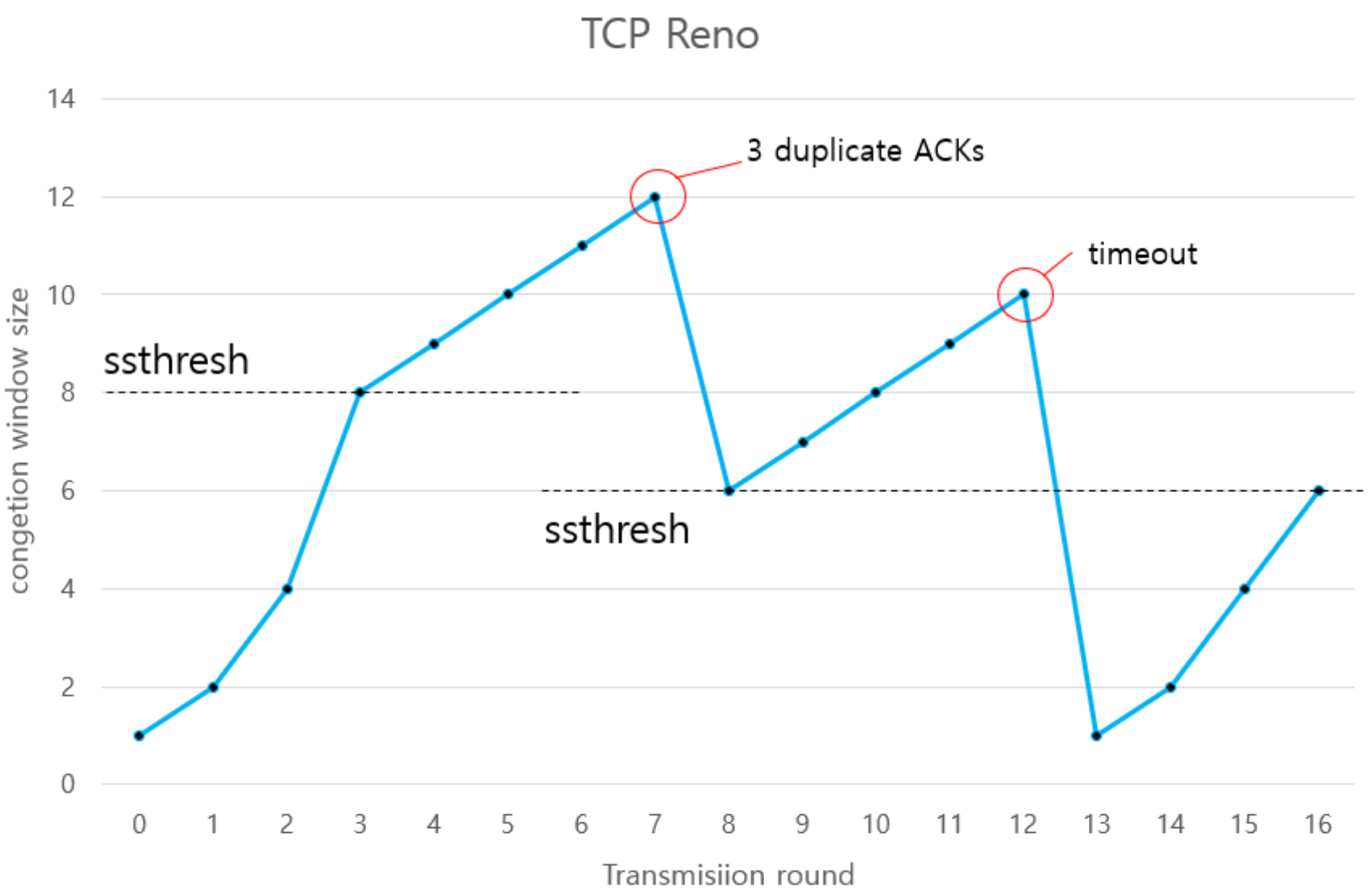
- The initial window size is 1 MSS.
- Until the threshold is reached, Slow Start is used (the window size doubles each round-trip time).
- After reaching the threshold, the AIMD approach is used (the window size increases by 1 MSS each round-trip time).
- When 3 duplicate ACKs or a timeout occurs, the threshold is set to half of the current window size, and the window size is reduced to 1 MSS.


TCP Tahoe
3 duplicate ACKs or timeout
ssthresh
ssthresh
congetion window size
Transmisiion round
0
2
4
6
8
10
12
14
0
1
2
3
4
5
6
7
8
9
10
11
12
13
14
15

TCP Reno

TCP Reno is similar to TCP Tahoe, but it differs in that it distinguishes between 3 duplicate ACKs and a timeout.

- The initial window size is 1 MSS.
- Until the threshold is reached, Slow Start is used (the window size doubles each round-trip time).
- After reaching the threshold, the AIMD approach is used (the window size increases by 1 MSS each round-trip time).
- When 3 duplicate ACKs are received, the window size is reduced to half, and the threshold is set to that value.
- When a timeout occurs, the window size is reduced to 1 MSS, but the threshold remains unchanged.


TCP Reno
3 duplicate ACKs
timeout
ssthresh
ssthresh
congetion window size
Transmisiion round
0
2
4
6
8
10
12
14
0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16

Reserved TCP Port Numbers

- Like UDP:
  - Static port bindings for commonly used services
    - Ports 0-1024 are reserved
  - Dynamic port bindings
    - Port numbers over 1024
- Port numbers for services accessible by both UDP and TCP usually match
  - ECHO (7)
  - TIME (37)

Reserved TCP Port Numbers

Port number:	Service:
0	Reserved
7	Echo
17	Quote of the day
21	FTP
23	TELNET
25	SMTP
37	Time
79	Finger
80	HTTP
119	NNTP

Summary

- **TCP Window** defines the flow of data between sender and receiver.
- **Window Size** indicates how much data the receiver can handle at a time.
- **Sliding Window** adjusts dynamically based on ACKs.
- **Flow Control** ensures the sender does not overwhelm the receiver's buffer.